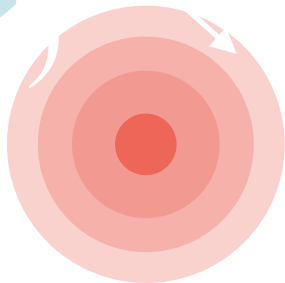
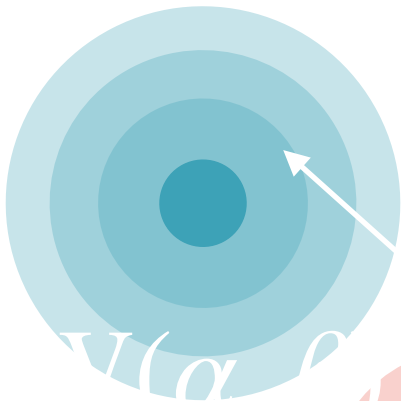
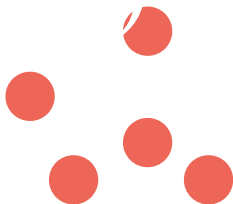
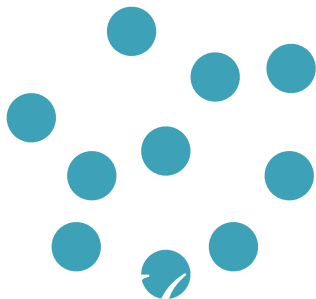


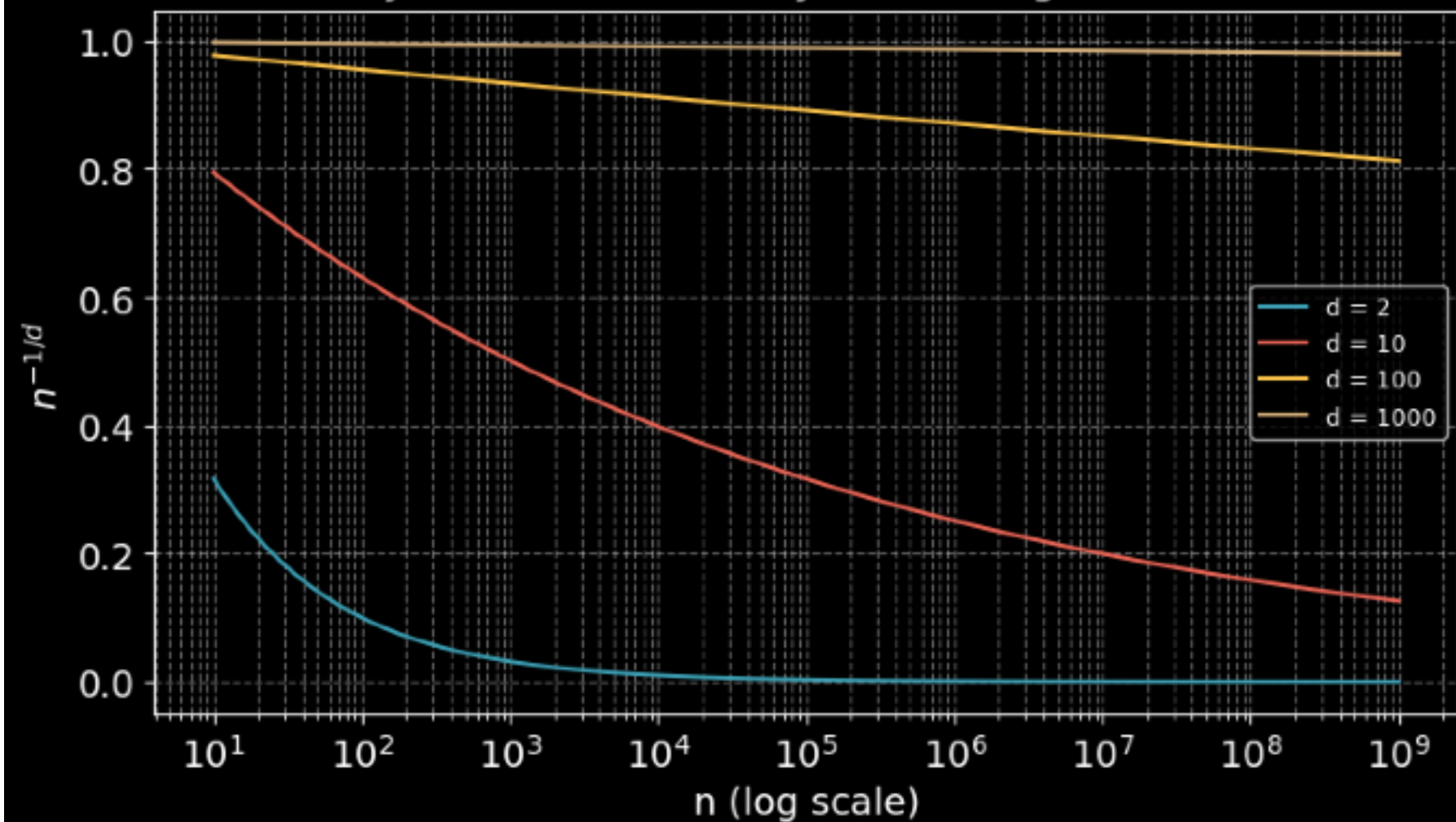
Sample complexity $\sim$ 'hardness' of estimating

Bad news



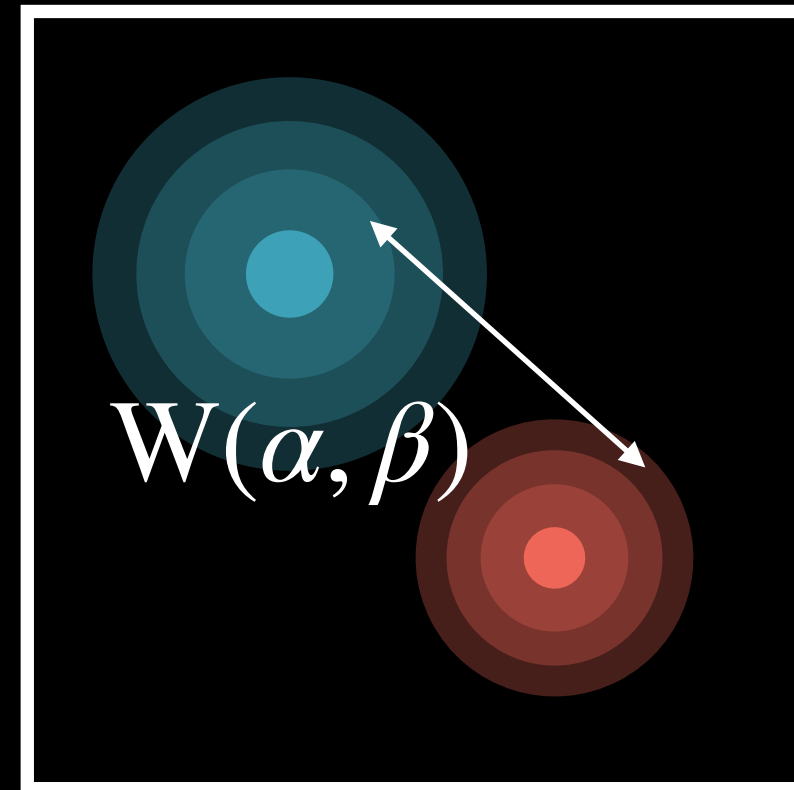


Decay of $n^{-1/d}$: extremely slow in high dimensions

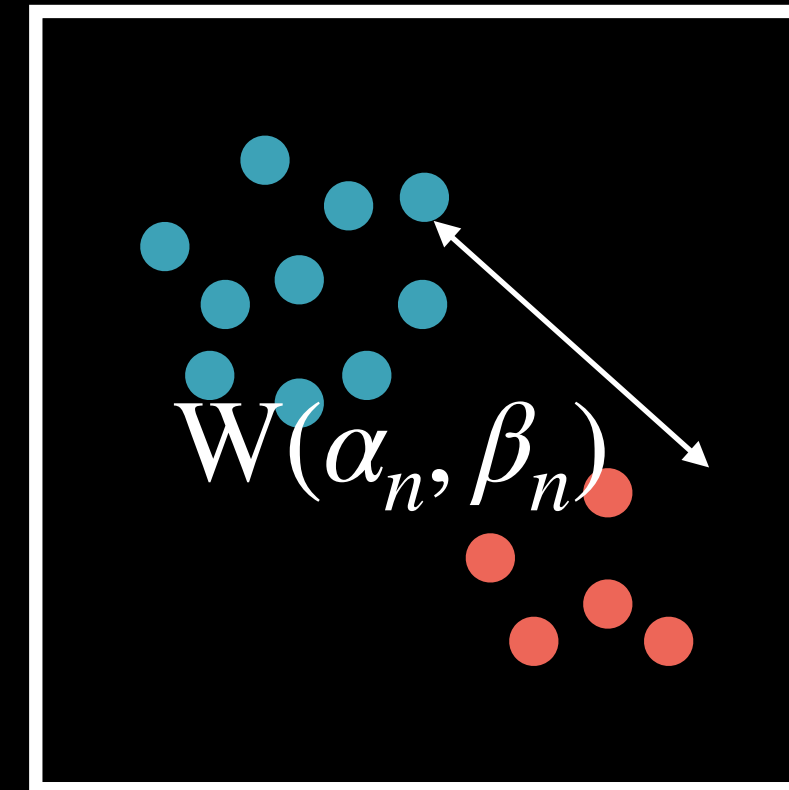


Side Note: Sample Complexity

If we estimate

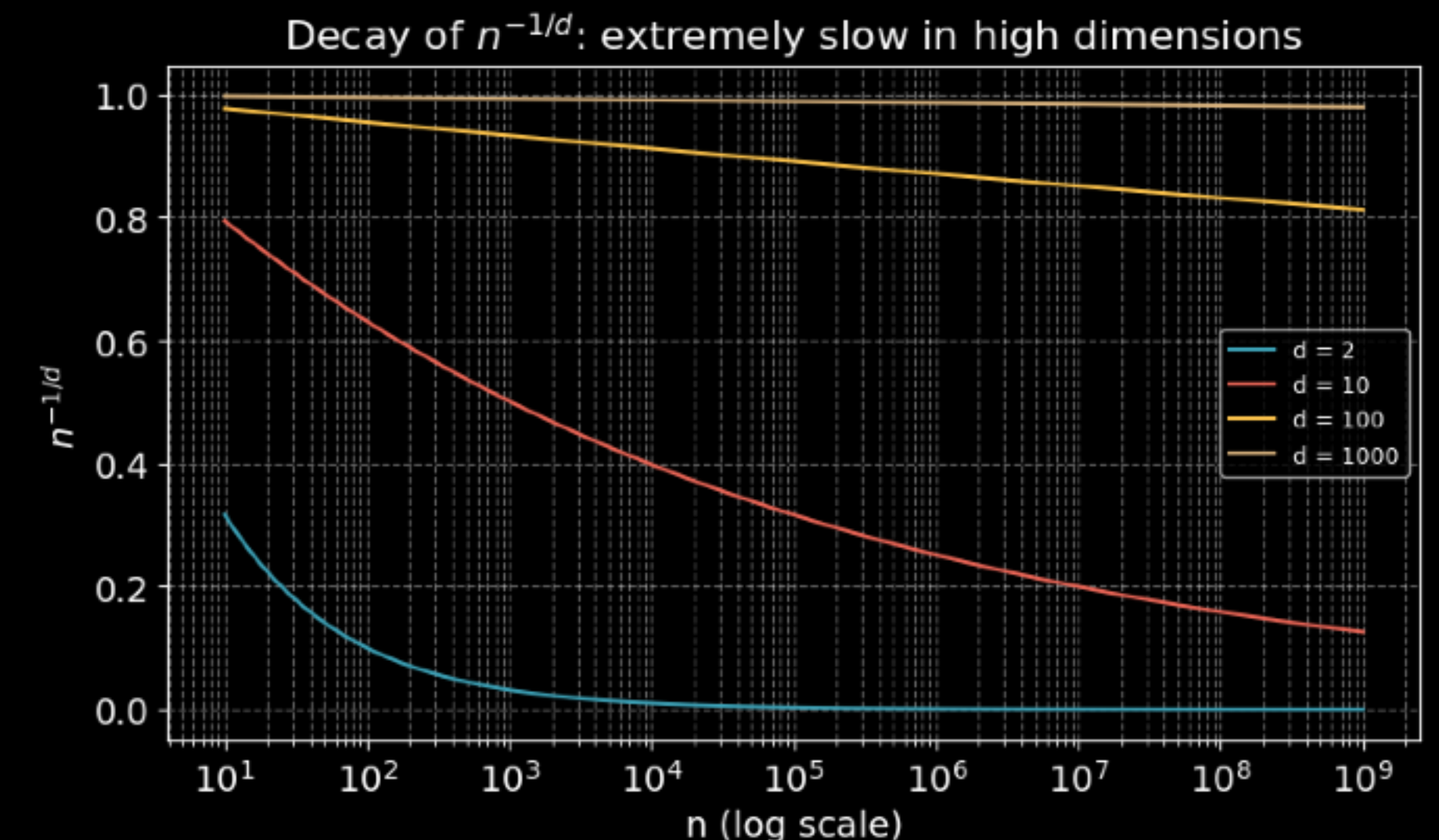


with



, how large should n be?

- **Intuition: Sample complexity** $\sim$ 'hardness' of estimating statistical quantity from (finite) data
- We're interested in $W_p(\hat{\alpha}_n, \hat{\beta}_n) \leftrightarrow W_p(\alpha, \beta)$
- **Bad news:** estimating OT is (statistically) hard!
- Indeed, $\mathbb{E}(|W_p(\hat{\alpha}_n, \hat{\beta}_n) - W_p(\alpha, \beta)|) = O(n^{-\frac{1}{d}})$
- An example of curse of dimensionality!



Sample Complexity of SD

- Classic OT: $\mathbb{E}(|W_p(\hat{\alpha}_n, \hat{\beta}_n) - W_p(\alpha, \beta)|) = O(n^{-\frac{1}{d}})$

